9.6. ČERENKOV DETECTORS

The Čerenkov effect described in section 7.4.3 can be used for detection of high energy β-radiation because the velocity of the nuclear particle must exceed the ratio $\mathbf{c}/n$, where n is the refractive index of the absorber.

$$E_{\beta\text{threshold}} = 0.511 \left[-1 + \left\{ 1 + \left(n^2 - 1 \right)^{-1} \right\}^{1/2} \right] \left(\text{MeV} \right) \tag{9.28}$$

The β-threshold energy in lucite ($n = 1.5$) is 0.17 MeV, so lucite and similar plastics are often used as particle absorbers in Čerenkov detectors. In order to detect the light emitted, PMTs are placed in the direction of the emitted light. There are many similarities between scintillation and Čerenkov detectors; however, the light pulse from the Čerenkov detector is faster, $\sim 10^{-10}$ s, but smaller than from scintillation detectors. The advantage of the Čerenkov detector is that aqueous or organic (uncoloured) solutions can be used without the need to add a scintillator, that soft β-emitters and γ-emitters give little or no signal, and that the angle of emitted light reveals the velocity of the absorbed particles, cf. Eqn (7.18). At energies <10 MeV, only β-particles are detected.

9.7. MICROCHANNEL PLATE DETECTORS

A microchannel plate detector (MCP) is a planar component produced by cutting packages of sintered microtubes at an angle to their axis and used for detection and amplification of particles and radiation. The MCP consists of an insulating thin plate penetrated by very many holes and sandwished between two thin conducting surfaces which are also penetrated by the holes. In use, a high voltage is applied between the conducting surfaces. The MCP is a relative to the PMT-tube, as both intensify single particles or photons by the multiplication of electrons via secondary emission. However, a microchannel plate detector has many separate channels in parallel, each of which works like a small windowless PMT. Hence, it can also provide some spatial resolution. In general two types of MCP:s are in use. In one type the channels are straight and at an angle of about 8° relative to the surface. In another type two plates are sandwished with their hole-angles in a zig-zag arrangement, which yields a higher amplification. The channels have a diameter in the range 5—10 μm and are spaced about 15 μm apart. MCP-devices are used as detectors in mass spectrometers and as amplifiers in light-intensifying devices.

By using a very thin carbon foil positioned at an angle above the entrance side of the MCP, electrons released when charged particles transit through the carbon foil with very small energy loss can be collected, amplified and detected. This gives an electric signal from any passing particle or bunch of particles which can be used for timing purposes.

9.8. ELECTRONICS FOR PULSE COUNTING

A wide variety of counting systems has been developed for various purposes. Equipment is often built as NIM (Nuclear Instrument Module standard) or CAMAC (Computer Automated Measurement And Control standard) modules, which fit into standard bins (or crates) containing power supplies and some inter-module connections. Cable connectors are also largely standardized. This facilitates combinations of bias supplies, amplifiers, discriminators, SCAs, ADCs, counters, and other circuitry to fit any need as well as their connection to computers. Several types of counting systems are also designed as plug-in cards to AT-type PC:s. Some non-standard units, e.g. portable instruments, have their own power supply, main amplifier, counter or rate-meter, etc. Today, it is common that even though a detector system comprises all the components above they are not always separate units but integrated in the detector which in turn is connected to a computer using e.g. a USB cable.

9.8.1. Preamplifiers

The purpose of a preamplifier may be twofold. First of all it should increase the energy in the detector signal to such a level that it can drive a reasonable length of low impedance coaxial cable properly terminated at the other end with only a small loss in pulse amplitude. When the detector produces a very weak signal it may be advantageous to amplify that signal to a level where external noise becomes negligible. However, in some applications the detector signal is already large enough, the detector capacitance constant (e.g. a PMT) and *voltage sensitive preamplifiers* with unit gain are used, Figure 9.19(a).

A 0.5 MeV γ absorbed in a germanium crystal will only produce a charge of 2.7×10^{-14} C, Eqn (9.2). Moreover, the detector capacitance may change with operating conditions which make voltage sensitive preamplifiers less suitable and charge sensitive preamplifiers are preferred. The weak charge signal has to be integrated and converted to a voltage pulse by the preamplifier without adding too much noise. Figure 9.19(b) shows a typical input stage for a preamplifier used with germanium γ-ray detectors.

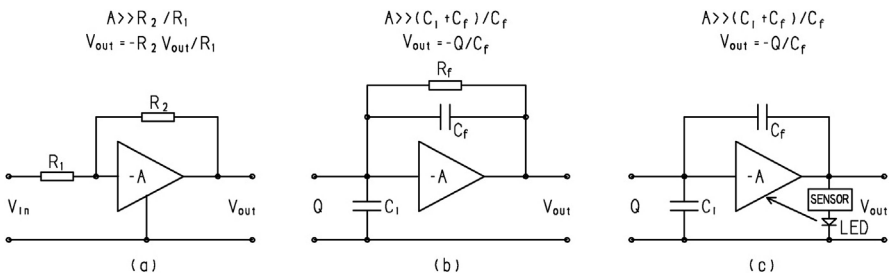


Figure 9.19 Voltage sensitive (a), charge sensitive resistive feedback (b), and pulsed optical feedback (c) preamplifiers. A is amplifier gain.